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Practical no 3.

Aim:- install Jcrypt tool and perform encryption and decryption of text or files using the specific cryptographic algorithm.

Resources Required : Hardware - computer system.
software - Jcrypt tool.

objectives

- To install the Jcrypt tool for cryptographic operations.
- To perform encryption and decryption of text or files.

Theory:

Cryptography

Cryptography is the technique of securing information by converting it into an unreadable format to prevent unauthorized access.

plaintext → original readable data.

ciphertext → encrypted unreadable data.

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Types of Cryption.

- 1) Symmetric key cryptography.
 - Same Key used for encryption and decryption.

ex:- AES, DES.

- 2) Asymmetric key Cryptography.
 - Uses two keys (Public & Private)
 - Example: RSA.

AES Algorithm

- AES (Advanced encryption standard) is a symmetric encryption algorithm.
- works on:

- 128-bit block size
- Key sizes: 128, 192, 256 bits.

provides high security and is widely used.

Jcrypt Tool

- Jcrypt is a Java based cryptographic tool

it allows:

- encryption and decryption
- Files and text processing
- selection of algorithm like AES.

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Requirements :

- JDK installed
- Jcrypt tool (.jar file)
- command Prompt.

Steps to install Jcrypt toolstep 1 : Install Java

- 1) download java from official website
- 2) Install JDK
- 3) verify installation : java version.

step 2 : Download Jcrypt Tool

- Download Jcrypt.jar file from trusted source.
- Save it in a folder.

Step 3 : Run Jcrypt Tool

- open command Prompt and navigate :
cd c:\Jcrypt.

Run the tool :java -jar Jcrypt.jarEncryption of Text

- 1) open Jcrypt tool
- 2) select : Algorithm : AES
- 3) Enter : plaintext : Hello 123

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• key : secretkey

4) click encrypt.

Decryption of Text

1) copy the ciphertext

2) Enter same key : Secretkey.

3) Click Decrypt.

output: original text (Hello123)

Advantages : provides storing data security.

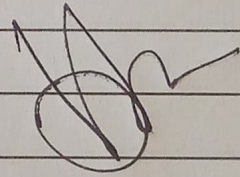
• Easy to use interface.

• Supports multiple algorithm.

Applications : secure communication.

Banking system.

conclusion:- Thus, I successfully installed and used the Jcrypt tool. performed encryption and decryption using AES Algorithm.



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Practical no 4.

Aim:- To install an AdBlocker extension and study its effect on blocking online advertisement

Resource Required:- computer System.

software - Ad blocker Extension

Objective -

- To install an Ad Blocker extension in a web browser.
- To study how ad blocker prevent online advertisement.

Theory

- online Advertisement.

online advertisements are promotional content displayed on websites in various forms :

- 1) Banner ads
- 2) pop ups
- 3) video ads.
- 4) sponsored links

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Ad Blocker :

An Ad blocker is a browser extension that :

- Detects and blocks ads on web pages.
- prevents tracking scripts
- improves browsing speed and privacy.

How Ad Blockers works :Ad blockers use :1) Filter lists

- predefined rules (eg. Easylist)
- Block URLs related to ads.

2) DNS blocking

- stops communication with ad servers.

3) Element Hiding

- removes ad sections from webpages.

Example tools :

- Adblock Plus
- uBlock Origin
- Ad Guard

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Requirements

- computer with internet connection.
- web browser : google chrome.

Steps to install Ad BlockerStep 1: open Browser.

- launch Google chrome.

Step 2: open extensions store.

- go to chrome web store.
- search for dBlock origin.

Step 3: Install extension

- click Add to chrome.
- confirm installation.

Step 4: verify installation.

- Extension icon appears in toolbar.
- click icon to check status.

observation

parameter	without Ad blocker	with ad blocker
number of ads	high	very low

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pages load time.	slow	Faster
Data usage	high	reduced
privacy protection	low	high.

Example :

website : news site

• Before installation.

• 10-15 ads visible

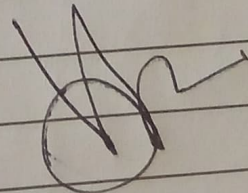
• Auto-playing video ads.

• After installation.

• no ads.

• Faster loading.

conclusion:- Thus, successfully installed and ad blocker extension (uBlock origin) observed significant reduction in online advertisements.



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Practical no 5

Aim:- study and demonstrate short url creation tools like tinyurl, checkshort url etc.

Objective: To study the concept of URL shortening.

- To demonstrate the use of tools like TinyURL and checkshortURL.

Theory:

URL shortening is a technique used to convert a long URL into a shorter, manageable link.

Example :

- original URL:

<https://www.example.com/products/category/item/2345>

- short URL:

<https://tinyurl.com/x4z123>

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need for URL shortening?

- Easy sharing on social media
- Better readability
- saves space.
- useful in marketing and analytics.

Short URL Tools:

- Tiny URL
- checkShortURL
- Bitly.

Working of URL shortening:

Process:

- 1) user enters a long URL.
- 2) server generates a unique short code.
- 3) stores mapping in database.
- 4) provides short URL.

Redirection Process

user clicks short URL → server checks database → Redirects to original URL.

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creating short URL using TinyURL:

- 1) open browser
- 2) Go to TinyURL
- 3) Enter long URL
- 4) click "make TinyURL"
- 5) Copy generated short link.

ex:- Input URL:

<https://www.wikipedia.org/wiki/cryptography>

output:

<https://tinyurl.com/abc123>Verifying short URL using checkshortURL

- 1) open checkshortURL
- 2) paste short URL
3. click Expand
- 4) view.
 - original URL
 - safety information.

Short URL creation

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long URL → Hash / code Generation → stored in database → short URL created.

URL Redirection.

Short URL → server lookup → original URL
Redirect user.

observation.

parameter	long URL	short URL.
length	very long	short & manage
sharing	Difficult	Easy.
Tracking	limited	Advance

Advantages : Easy to share links.
Improves user experience.

conclusion :- successfully studied and demonstrated URL shortening tools.
understood both advantages and
disadvantages of short URLs.

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Practical no 6

Aim:- To install spamfighter and study how it filters spam email from the in box.

Objectives:

- To install SPAMfighter email filtering software.
- To study how spam emails are detected and filtered.

Resources Required:- computer system.
Software - spamfighter.

Theory:

Spam Email:

spam emails are unsolicited or unwanted messages sent in bulk via email. These emails are usually:

- Advertisements.
- phishing Attempts.
- malware - containing messages.

problems caused by spam:

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- inbox clutter.
- security threats (viruses, phishing)
- wastage of time and bandwidth.

SPAMfighter

SPAMfighter is an anti spam tool that:

- integrates with email clients.
- Automatically detects and filters spam emails.
- uses community-based filtering.

working Principle of SPAMfighter:

1) community - Based Filtering.

- when one user marks an email as spam:
- it is reported to SPAMfighter servers.
- All users benefit from this shared database.

2) Blacklist and whitelist.

Blacklist → Blocks known spam senders.
whitelist → Allows trusted senders.

3) pattern matching

- detects spam based on.

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- keywords
- suspicious Formatting.

Heuristics

- learns from user behavior.
- Improves detection accuracy over time

Header Analysis

- checks sender details.
- identifies fake or spoofed emails.

Steps to install SPAMfighter

- 1) Download software.
 - go to official SPAMfighter website.
 - Download setup file.
- 2) Install Application.
 - Run setup file
 - Follow installation wizard.
 - Accept license agreement.
- 3) Restart Email client.
 - open microsoft outlook.
 - SPAMfighter toolbar appears.

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4) configure settings.

- enable spam protection.
- update filters.

After Installing SPAMfighter:

1) enable SPAMfighter.

2) check inbox

3) observe:

- spam emails moved to spam folder
- Reduced unwanted emails.

marking Emails:

- click "spam" → moves email to spam folder.

- click "not spam" → restores emails.

Detailed flow

1) Email arrives.

2) SPAMfighter scans content & sender.

3) compares with database.

a) classifies as:

- spam
- legitimate.

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Advantages:

- Automatically filters spam.
- Improves email security.
- reduces phishing risks.
- easy to use.

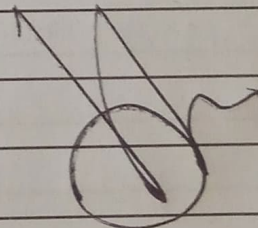
Limitations:

- may blocks important emails.
- requires internet for updates.

Application:

- personal email protection.
- corporate emails system.

conclusion: successfully installed SPAMfighter observed effective filtering of spam emails.



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Practical no 7

Aim :- To use Spambully to identify and remove spam and phishing emails from an email account.

Objective :

- To study the concept of spam and phishing emails.
- To install and use Spambully email filtering software.

Resource Required : computer system.

Theory :Spam Email :

Spam emails are unsolicited bulk messages sent to users without their consent. These emails usually contain:

- advertisements.
- fake offers.
- malicious links.

Phishing Emails :

phishing emails are fraudulent emails designed to:

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- steal sensitive information (password, OTPs, bank details)
 - impersonate trusted organization.
- ex:- fake bank login page.
your account is blocked message.

SpamBully :

Spambully is an antispam tool that:

- works with email clients like microsoft outlook.
- uses intelligent filtering technique.
- detects both spam and phishing emails.

working principle of spamBully :

① Bayesian filtering.

- uses probability based filtering.
- learns from user actions.
- classifies emails as spam or not.

② challenge - response system.

- unknown senders must verify themselves.
- prevents automated spam bots.

③ header analysis.

- checks sender Authenticity.
- detects spoofed email addresses.

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5) link analysis :

- scans URLs inside emails.
- identifies fake or malicious links.

Steps to install spamBully.

1) Download software.

- visit official spamBully website.
- download installation file.

2) Install Application.

1) Run Setup file.

2) follow installation wizard.

3) Accept license agreement.

4) select email client integration.

3) launch Email client.

- open Microsoft outlook.
- spamBully toolbar appears.

4) configure settings.

- enable spam filtering.
- updates rules and database.

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Detailed Flow.

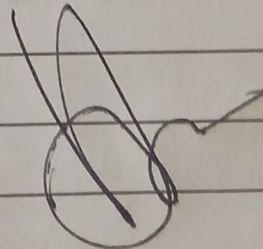
- 1) Email arrives.
- 2) SpamBully scans:
 - content
 - sender
 - links.
- 3) Applies filtering techniques.
 - a) classifies emails.
 - spam
 - phishing.
 - legitimate.

Advantages:

- ① Detects both spam and phishing.
- ② Improves email security.
- ③ Reduces unwanted emails.
- ④ Learns from user behavior.

Conclusion:- Successfully used SpamBully to filter emails.

Identified and removed spam and phishing emails effectively.



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Practical no 8.

Aim. To study and implement user authentication using time based one time password totP google authenticator.

Resource Required: computer system
operating system.

objective:- To study multifactor authentication using TOTP.

- To implement Time based one time Password TOTP authentication.

Theory:

user Authentication :- user authentication is the process of verifying the identify of a user before granting access to a system.

Types of Authentication Factors

- 1) knowledge factor - Password pin
- 2) possession factor - OTP, smart card
- 3) Biometric factor - Fingerprint.

Multifactor Authentication:

MFA combines two or more Authentication factor.

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Example : Password + OTP

- Improves security by adding an extra layer of protection.

One time password :

OTP is a password that :

- Is valid for only one login session
- Expires after a short time.

Time based one time Password :

TOTP is a type of OTP that

- changes every 30 second.
- is generated using:
secret key
current time.

google Authenticator

google Authenticator is a mobile Application that :

- generates TOTP codes.
- works Offline.
- provides secure authentication.

1) server generates a Secret Key.

2) user scans QR code in google authenticator

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3) APP generates OTP during login.

4) Server verifies OTP.

Practical implement:

Install Required library.

• pip install pyotp qrcode.

generate secret code and QR

`import pyotp``import qrcode``secret = pyotp.random_base32()``print("secret key:", secret)``TOTP = pyotp.TOTP(secret)``uri = TOTP.provisioning_uri(name="user@example.com", issuer_name="myApp")``img = qrcode.make(uri)``img.save("qrcode.png")`

STEPS google Authenticator

open google Authenticator

click + Add account

scan QR code generated.

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Verify OTP

```
OTP = input("Enter OTP: ")
```

```
if totp.verify(otp):
```

```
    print("Authentication successful")
```

```
else:
```

```
    print("Invalid OTP")
```

Advantages : high security

works offline

prevents password theft attacks.

Easy to implement.

Application : Banking systems

Email Authentication.

cloud service

secure login system.

conclusion: successfully implemented TOTP authentication using google authenticator. learned how OTP enhances security in login systems.

